Adithya Hegde

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github.com/TheMonocledHamster

Education

Georgia Institute of Technology, MS in Computer Science - Computing Systems — Expected May 2028

The National Institute of Engineering, BE in Computer Science and Engineering — Jun 2023

Experience

Software Engineer II, JPMorgan Chase & Co. — Jan 2025 – Jul 2026

- Designed and delivered a new distributed data processing platform using PySpark and Java, replacing legacy systems with full ownership of architecture, implementation, and production rollout.
- Redesigned platform execution to eliminate Databricks-DynamoDB state synchronization, removing a fragile distributed dependency and improving throughput by up to 80%.
- Reduced runtime of a ∼60-operation Spark DAG processing up to 10M records from ∼10 minutes to ∼1.5 minutes (85%) through logical-plan optimization and data-dependent DAG construction.
- Tuned Spark workload partitioning and task parallelism to reduce shuffle overhead, cutting cluster compute costs by ∼40% while maintaining execution performance.
- Optimized table partitioning using Spark execution plans and Databricks query profiles, increasing data skipping from 65% to 98% and reducing dashboard refresh latency from ∼5 minutes to ∼10 seconds.

Software Engineer I, JPMorgan Chase & Co. — Jul 2023 – Dec 2024

- Profiled memory growth of a Polars-based transformation running in AWS Lambda's constrained runtime; introduced lazy evaluation and explicit object lifecycle control, reducing peak memory allocation by 60% and stabilizing execution latency.
- Brought up a multi-node ingestion system on Kubernetes (EKS + NiFi), implemented Airflow orchestration for distributed workflows, and operated cross-service data movement pipelines in production.
- Accelerated promotion to Software Engineer II

Projects

XDBC: Cross-DBMS Data Transfer | GitHub | Go, PostgreSQL — Fall 2026

- Co-developing an end-to-end XDBC data transfer system in Go, including client/server pipelines, wire-protocol messaging, buffer management, and PostgreSQL integration.
- Reproducing XDBC data transfer benchmarks against PostgreSQL FDW baselines to evaluate throughput and transfer overhead.

Publications

COUNSEL - Cloud Resource Configuration Management using Deep RL — May 2023

Proposed a Deep RL framework for cloud resource configuration using queueing-based service-chain modeling.
Adithya Hegde, Sameer G. Kulkarni, Abhinandan S. Prasad
10.1109/ccgrid57682.2023.00035 (**CCGRID 2023**)

Skills

Languages: Python, Java, Go, SQL; Familiarity with CPP, Rust

Data Systems: Apache Spark, Databricks, PostgreSQL, Airflow, Pytorch

Infrastructure: Kubernetes, AWS, Linux